

2. All of the eigenvalues of A have magnitude less than or equal to one, and those with magnitude equal to one have a geometric multiplicity equal to their algebraic multiplicity. That is, the Jordan blocks that are associated with the eigenvalues that have magnitude equal to one are first order.

Theorem 8 *A linear discrete-time, time-invariant system is asymptotically stable if and only if all of the eigenvalues of A have magnitude less than one.*

1.7 CONTROLLABILITY AND OBSERVABILITY

The concepts of controllability and observability are fundamental to modern control theory. These concepts define how well we can control a system (i.e., drive the state to a desired value) and how well we can observe a system (i.e., determine the initial conditions after measuring the outputs). These concepts are also important to some of the theoretical results related to optimal state estimation that we will encounter later in this book.

1.7.1 Controllability

The following definitions and theorems give rigorous definitions for controllability for linear systems in the both the continuous-time and discrete-time cases.

Definition 5 *A continuous-time system is controllable if for any initial state $x(0)$ and any final time $t > 0$ there exists a control that transfers the state to any desired value at time t .*

Definition 6 *A discrete-time system is controllable if for any initial state x_0 and some final time k there exists a control that transfers the state to any desired value at time k .*

Note the controllability definition in the continuous-time case is much more demanding than the definition in the discrete-time case. In the continuous-time case, the existence of a control is required for *any* final time. In the discrete-time case, the existence of a control is required for *some* final time. In both cases, controllability is independent of the output equation.

There are several tests for controllability. The following equivalent theorems can be used to test for the controllability of continuous linear time-invariant systems.

Theorem 9 *The n -state⁶ continuous linear time-invariant system $\dot{x} = Ax + Bu$ has the controllability matrix P defined by*

$$P = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \quad (1.138)$$

The system is controllable if and only if $\rho(P) = n$.

Theorem 10 *The n -state continuous linear time-invariant system $\dot{x} = Ax + Bu$ is controllable if and only if the controllability grammian defined by*

$$\int_0^t e^{A\tau} B B^T e^{A^T \tau} d\tau \quad (1.139)$$

⁶The notation *n-state system* indicates a system that has n elements in its state variable x .

is positive definite for some $t \in (0, \infty)$.

Theorem 11 *The n -state continuous linear time-invariant system $\dot{x} = Ax + Bu$ is controllable if and only if the differential Lyapunov equation*

$$\begin{aligned} W(0) &= 0 \\ \dot{W} &= WA^T + AW + BB^T \end{aligned} \quad (1.140)$$

has a positive definite solution $W(t)$ for some $t \in (0, \infty)$. This is also called a Sylvester equation.

Similar to the continuous-time case, the following equivalent theorems can be used to test for the controllability of discrete linear time-invariant systems.

Theorem 12 *The n -state discrete linear time-invariant system $x_k = Fx_{k-1} + Gu_{k-1}$ has the controllability matrix P defined by*

$$P = \begin{bmatrix} G & FG & \dots & F^{n-1}G \end{bmatrix} \quad (1.141)$$

The system is controllable if and only if $\rho(P) = n$.

Theorem 13 *The n -state discrete linear time-invariant system $x_k = Fx_{k-1} + Gu_{k-1}$ is controllable if and only if the controllability grammian defined by*

$$\sum_{i=0}^k A^{k-i}BB^T(A^T)^{k-i} \quad (1.142)$$

is positive definite for some $k \in (0, \infty)$.

Theorem 14 *The n -state discrete linear time-invariant system $x_k = Fx_{k-1} + Gu_{k-1}$ is controllable if and only if the difference Lyapunov equation*

$$\begin{aligned} W_0 &= 0 \\ W_{i+1} &= FW_iF^T + GG^T \end{aligned} \quad (1.143)$$

has a positive definite solution W_k for some $k \in (0, \infty)$. This is also called a Stein equation.

Note that Theorems 9 and 12 give identical tests for controllability for both continuous-time and discrete-time systems. In general, these are the simplest controllability tests. Controllability tests for time-varying linear systems can be obtained by generalizing the above theorems. Controllability for nonlinear systems is much more difficult to formalize.

■ EXAMPLE 1.8

The RLC circuit of Figure 1.4 has the system description

$$\begin{bmatrix} \dot{v}_C \\ \dot{i}_L \end{bmatrix} = \begin{bmatrix} -2/RC & 1/C \\ -1/L & 0 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 1/RC \\ 1/L \end{bmatrix} u \quad (1.144)$$

where v_C is the voltage across the capacitor, i_L is the current through the inductor, and u is the applied voltage. We will use Theorem 9 to determine the conditions under which this system is controllable. The controllability matrix is computed as

$$\begin{aligned} P &= \begin{bmatrix} B & AB \end{bmatrix} \\ &= \begin{bmatrix} 1/RC & 1/LC - 2/R^2C^2 \\ 1/L & -1/RLC \end{bmatrix} \end{aligned} \quad (1.145)$$

From this we can compute the determinant of P as

$$|P| = 1/R^2LC^2 - 1/L^2C \quad (1.146)$$

The determinant of P is 0 only if $R = \sqrt{L/C}$. So the system is controllable unless $R = \sqrt{L/C}$. It would be very difficult to obtain this result from Theorems 10 and 11.

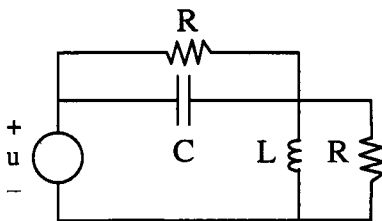


Figure 1.4 RLC circuit for Example 1.8.

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1.7.2 Observability

The following definitions and theorems give rigorous definitions for observability for linear systems in both the continuous-time and discrete-time cases.

Definition 7 A continuous-time system is observable if for any initial state $x(0)$ and any final time $t > 0$ the initial state $x(0)$ can be uniquely determined by knowledge of the input $u(\tau)$ and output $y(\tau)$ for all $\tau \in [0, t]$.

Definition 8 A discrete-time system is observable if for any initial state x_0 and some final time k the initial state x_0 can be uniquely determined by knowledge of the input u_i and output y_i for all $i \in [0, k]$.

Note the observability definition in the continuous-time case is much more demanding than the definition in the discrete-time case. In the continuous-time case, the initial state must be able to be determined at *any* final time. In the discrete-time case, the initial state must be able to be determined at *some* final time. If a system is observable then the initial state can be determined, and if the initial state can be determined then all states between the initial and final times can be determined.

There are several tests for controllability. The following equivalent theorems can be used to test for the controllability of continuous linear time-invariant systems.

Theorem 15 *The n -state continuous linear time-invariant system*

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx\end{aligned}\tag{1.147}$$

has the observability matrix Q defined by

$$Q = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}\tag{1.148}$$

The system is observable if and only if $\rho(Q) = n$.

Theorem 16 *The n -state continuous linear time-invariant system*

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx\end{aligned}\tag{1.149}$$

is observable if and only if the observability grammian defined by

$$\int_0^t e^{A^T \tau} C^T C e^{A \tau} d\tau\tag{1.150}$$

is positive definite for some $t \in (0, \infty)$.

Theorem 17 *The n -state continuous linear time-invariant system*

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx\end{aligned}\tag{1.151}$$

is observable if and only if the differential Lyapunov equation

$$\begin{aligned}W(t) &= 0 \\ -\dot{W} &= WA + A^T W + C^T C\end{aligned}\tag{1.152}$$

has a positive definite solution $W(\tau)$ for some $\tau \in (0, t)$. This is also called a Sylvester equation.

Similar to the continuous-time case, the following equivalent theorems can be used to test for the observability of discrete linear time-invariant systems.

Theorem 18 *The n -state discrete linear time-invariant system*

$$\begin{aligned}x_k &= Fx_{k-1} + Gu_{k-1} \\ y_k &= Hx_k\end{aligned}\tag{1.153}$$

has the observability matrix Q defined by

$$Q = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{n-1} \end{bmatrix} \quad (1.154)$$

The system is observable if and only if $\rho(Q) = n$.

Theorem 19 The n -state discrete linear time-invariant system

$$\begin{aligned} x_k &= Fx_{k-1} + Gu_{k-1} \\ y_k &= Hx_k \end{aligned} \quad (1.155)$$

is observable if and only if the observability grammian defined by

$$\sum_{i=0}^k (F^T)^i H^T H F^i \quad (1.156)$$

is positive definite for some $k \in (0, \infty)$.

Theorem 20 The n -state discrete linear time-invariant system

$$\begin{aligned} x_k &= Fx_{k-1} + Gu_{k-1} \\ y_k &= Hx_k \end{aligned} \quad (1.157)$$

is observable if and only if the difference Lyapunov equation

$$\begin{aligned} W_k &= 0 \\ W_i &= F^T W_{i+1} F + H^T H \end{aligned} \quad (1.158)$$

has a positive definite solution W_0 for some $k \in (0, \infty)$. This is also called a Stein equation.

Note that Theorems 15 and 18 give identical tests for observability for both continuous-time and discrete-time systems. In general, these are the simplest observability tests. Observability tests for time-varying linear systems can be obtained by generalizing the above theorems. Observability for nonlinear systems is much more difficult to formalize.

■ EXAMPLE 1.9

The RLC circuit of Example 1.8 has the system description

$$\begin{aligned} \begin{bmatrix} \dot{v}_C \\ \dot{i}_L \end{bmatrix} &= \begin{bmatrix} -2/RC & 1/C \\ -1/L & 0 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 1/RC \\ 1/L \end{bmatrix} u \\ y &= \begin{bmatrix} -1 & 0 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} \end{aligned} \quad (1.159)$$

where v_C is the voltage across the capacitor, i_L is the current through the inductor, and u is the applied voltage. We will use Theorem 15 to determine

the conditions under which this system is observable. The observability matrix is computed as

$$Q = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 2/RC & -1/C \end{bmatrix} \quad (1.160)$$

The determinant of the observability matrix can be computed as

$$|Q| = 1/C \quad (1.161)$$

The determinant of Q is nonzero, so the system is observable. On the other hand, suppose that $R = L = C = 1$ and the output equation is

$$y = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} \quad (1.162)$$

Then the observability matrix can be computed as

$$\begin{aligned} Q &= \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \\ |Q| &= 0 \end{aligned} \quad (1.163)$$

So the system is unobservable. It would be very difficult to obtain this result from Theorems 16 and 17.

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1.7.3 Stabilizability and detectability

The concepts of stabilizability and detectability are closely related to controllability and observability, respectively. These concepts are also related to the modes of a system. The modes of a system are all of the decoupled states after the system is transformed into Jordan form. A system can be transformed into Jordan form as follows. Consider the system

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \quad (1.164)$$

First find the eigendata of the system matrix A . Suppose the eigenvectors are denoted as $v_1, \dots, v_n$. Create an $n \times n$ matrix M by augmenting the eigenvectors as follows.

$$M = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} \quad (1.165)$$

Define a new system as

$$\begin{aligned} \dot{\bar{x}} &= M^{-1}AM\bar{x} + M^{-1}B \\ &= \bar{A}\bar{x} + \bar{B}u \\ y &= CM\bar{x} + Du \\ &= \bar{C}\bar{x} + Du \end{aligned} \quad (1.166)$$

The new system is called the Jordan form representation of the original system. Note that the matrix M will always be invertible because the eigenvectors of a matrix can always be chosen to be linearly independent. The two systems of Equations (1.164) and (1.166) are called algebraically equivalent systems. This is because they have the same input and the same output (and therefore they have the same transfer function) but they have different states.

■ **EXAMPLE 1.10**

Consider the system

$$\begin{aligned}
 \dot{x} &= Ax + Bu \\
 &= \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u \\
 y &= Cx + Du \\
 &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x + 2u
 \end{aligned} \tag{1.167}$$

This system has the same transfer function as

$$\begin{aligned}
 \dot{\bar{x}} &= \bar{A}\bar{x} + \bar{B}u \\
 &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \bar{x} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u \\
 y &= \bar{C}\bar{x} + Du \\
 &= \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \bar{x} + 2u
 \end{aligned} \tag{1.168}$$

The eigenvector matrix of A is

$$\begin{aligned}
 M &= \begin{bmatrix} v_1 & v_2 & v_n \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -3 \end{bmatrix}
 \end{aligned} \tag{1.169}$$

Note the equivalences

$$\begin{aligned}
 \bar{A} &= M^{-1}AM \\
 \bar{B} &= M^{-1}B \\
 \bar{C} &= CM
 \end{aligned} \tag{1.170}$$

The Jordan form system has two decoupled modes. The first mode is

$$\begin{aligned}
 \dot{\bar{x}}_1 &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \bar{x}_1 + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \\
 y_1 &= \begin{bmatrix} 1 & 0 \end{bmatrix} \bar{x}_1
 \end{aligned} \tag{1.171}$$

The second mode is

$$\begin{aligned}
 \dot{\bar{x}}_2 &= -2\bar{x}_2 + 0u \\
 y_2 &= \bar{x}_2
 \end{aligned} \tag{1.172}$$

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Definition 9 *If a system is controllable or stable, then it is also stabilizable. If a system is uncontrollable or unstable, then it is stabilizable if its uncontrollable modes are stable.*